LOAD BALANCER
High Availability | Scalability | Performance
CLIENTS
SERVER POOL
HEALTHY
HEALTHY
HEALTHY
HIGH AVAILABILITY
SCALABILITY
PERFORMANCE
HEALTH CHECK
DISTRIBUTES TRAFFIC EVENLY
PREVENTS SERVER OVERLOAD
AUTOMATIC FAILOVER & RECOVERY
SSL/TLS TERMINATION
FLEXIBLE CONFIGURATION

📌INTRODUCTION

A **Load Balancer** is a service that distributes incoming network traffic across multiple servers (such as EC2 instances) to ensure high availability, reliability, and optimal performance. It prevents any single server from becoming overloaded, improves application response time, and automatically routes traffic only to healthy servers. In cloud environments like **Amazon Web Services (AWS)**, load balancers help build scalable, fault-tolerant, and highly available applications.

PREREQUISITES

Before you begin, ensure you have:

- An AWS account
- Basics Knowledge of AWS EC2 Security Groups, Target Groups and Load Balancing
- Internet Access to log into AWS Management Console

1. Security Group

"First, I created a Security Group. A Security Group acts as a virtual firewall that controls inbound and outbound traffic. I allowed HTTP (80) and HTTPS (443) traffic for users, and SSH (22) for secure administration."
Role: Protects the Load Balancer and EC2 instances by allowing only required traffic.

2. EC2 Instance

"Next, I launched EC2 instances. An EC2 instance is a virtual server in AWS where I deployed my web application. These instances process user requests."
Role: Hosts and runs the application.

3. Target Group

"After that, I created a Target Group and registered my EC2 instances. The Target Group performs health checks and ensures that the Load Balancer sends traffic only to healthy EC2 instances."
Role: Groups EC2 instances and monitors their health.

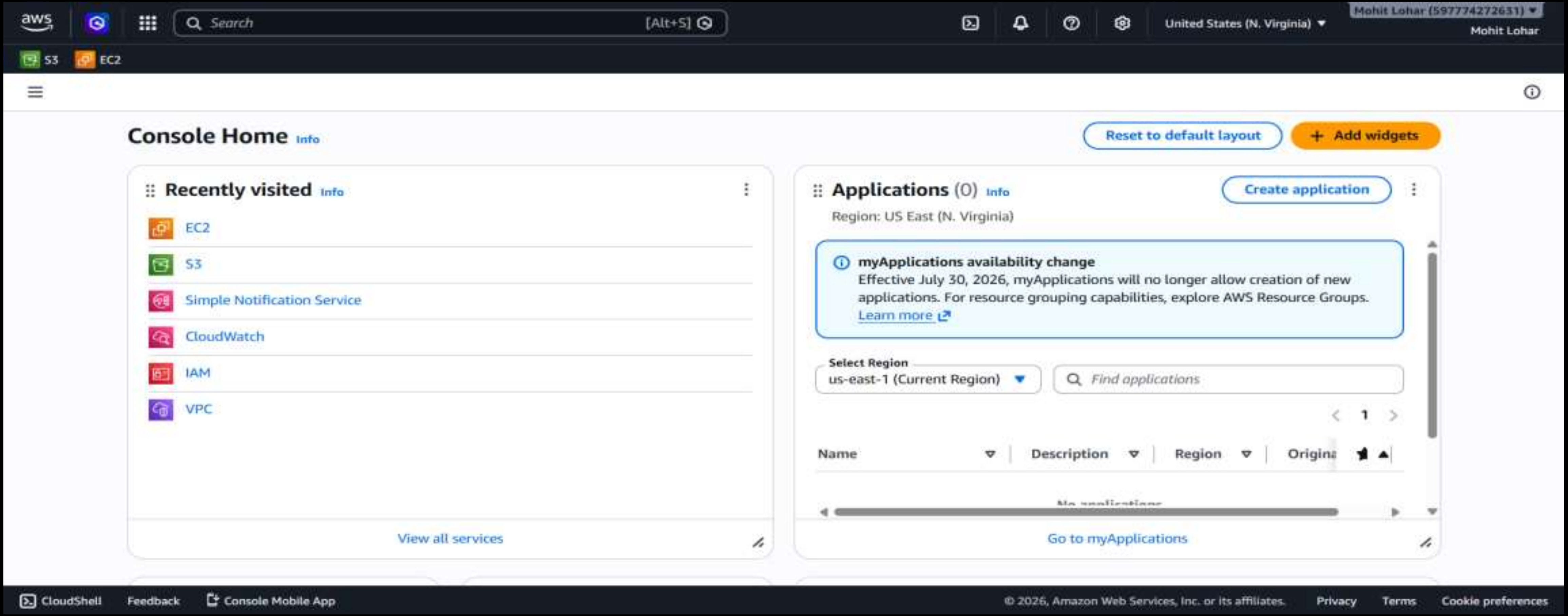
4. Load Balancer (ALB)

"Finally, I created an Application Load Balancer (ALB). I attached the Security Group, selected the VPC and subnets, and connected it to the Target Group. The Load Balancer receives incoming user requests and distributes them across the healthy EC2 instances."
Role: Distributes incoming traffic to healthy EC2 instances, ensuring high availability, scalability, and fault tolerance.

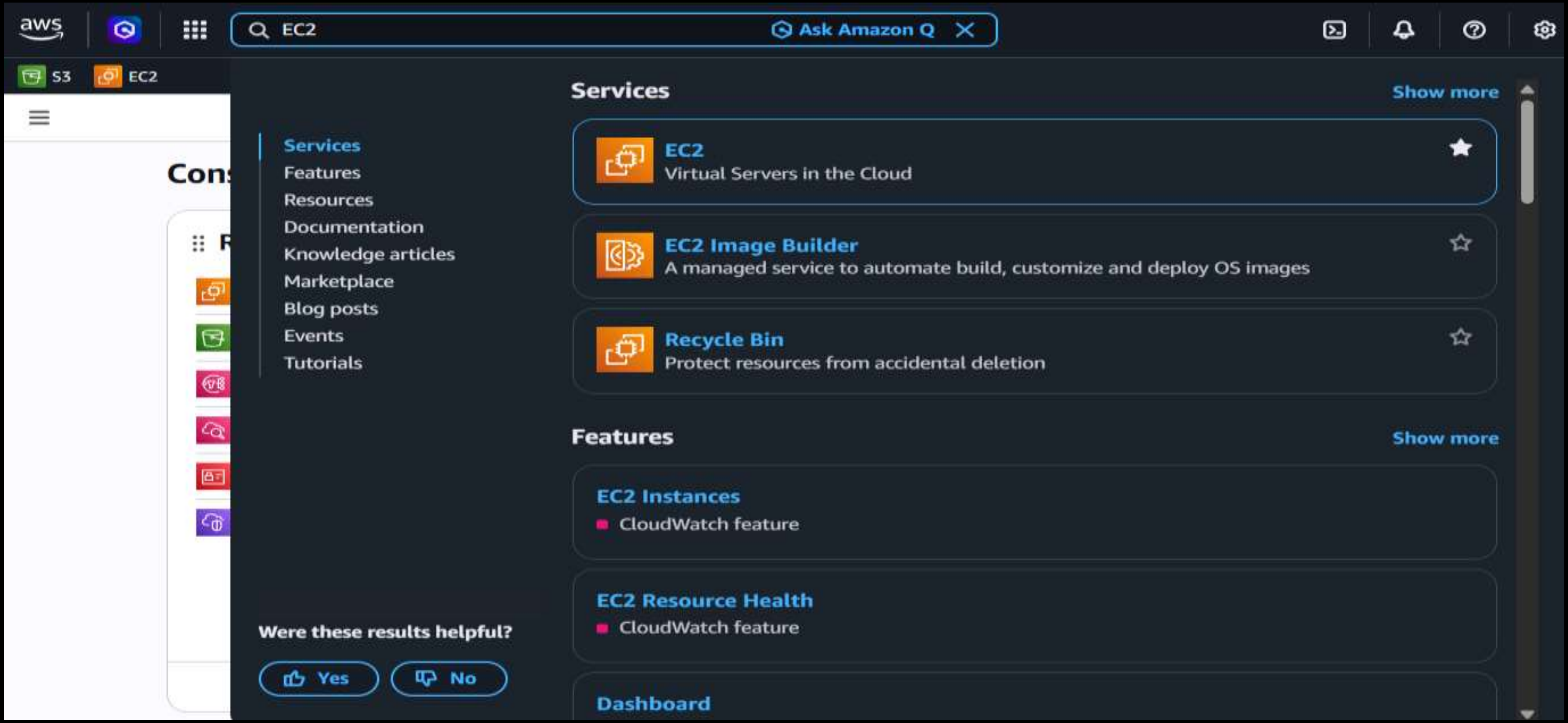
Steps to Set Up the Infrastructure

CONFIGURE SECURITY GROUP

1. Go to AWS Management Console.

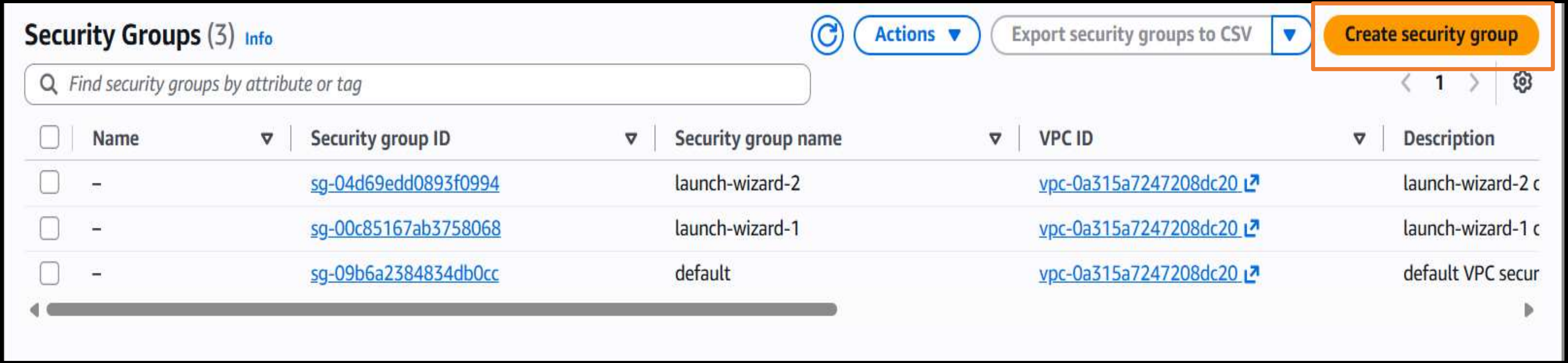


2. Search And Click EC2

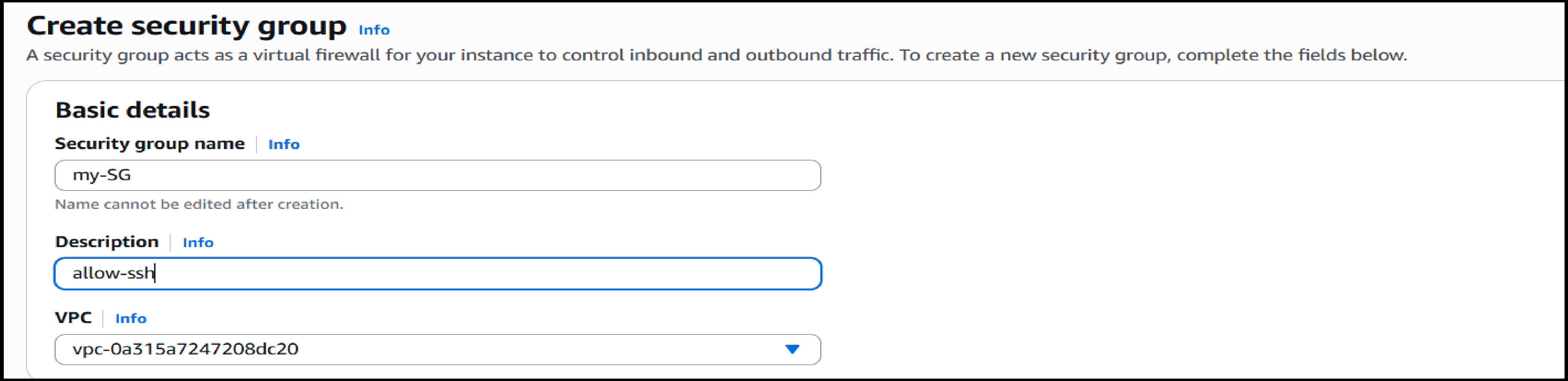


3. Click on Security Group

4. Click on **Create Security Group.**



5. Enter Basic Details like Name and description.



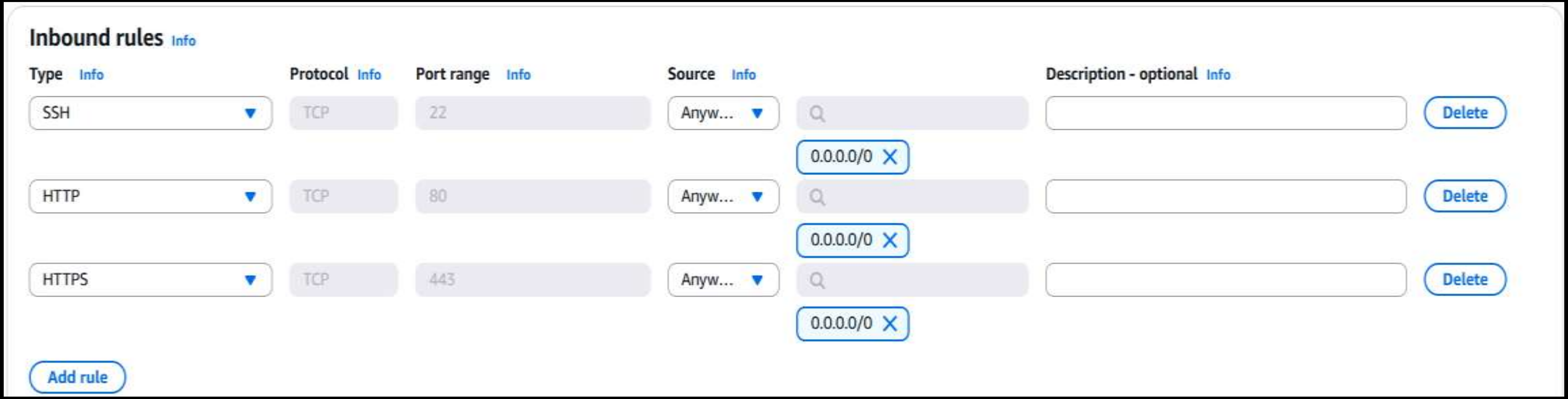
6. In Inbound Rules , add the following rules:

SSH (Port 22)

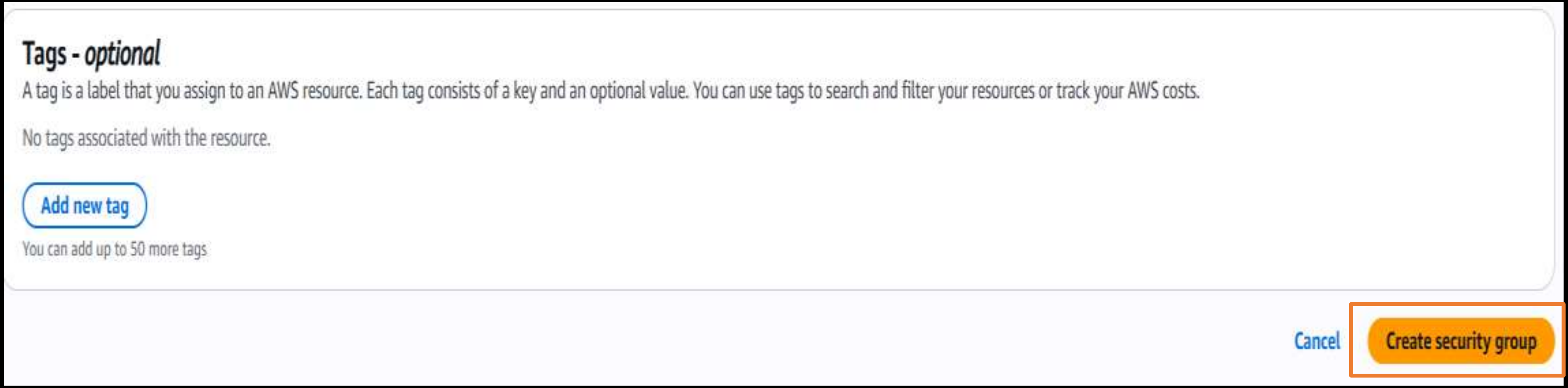
HTTP (Port 80)

HTTPS (Port 443)

Select Anywhere IPv4 as the source.

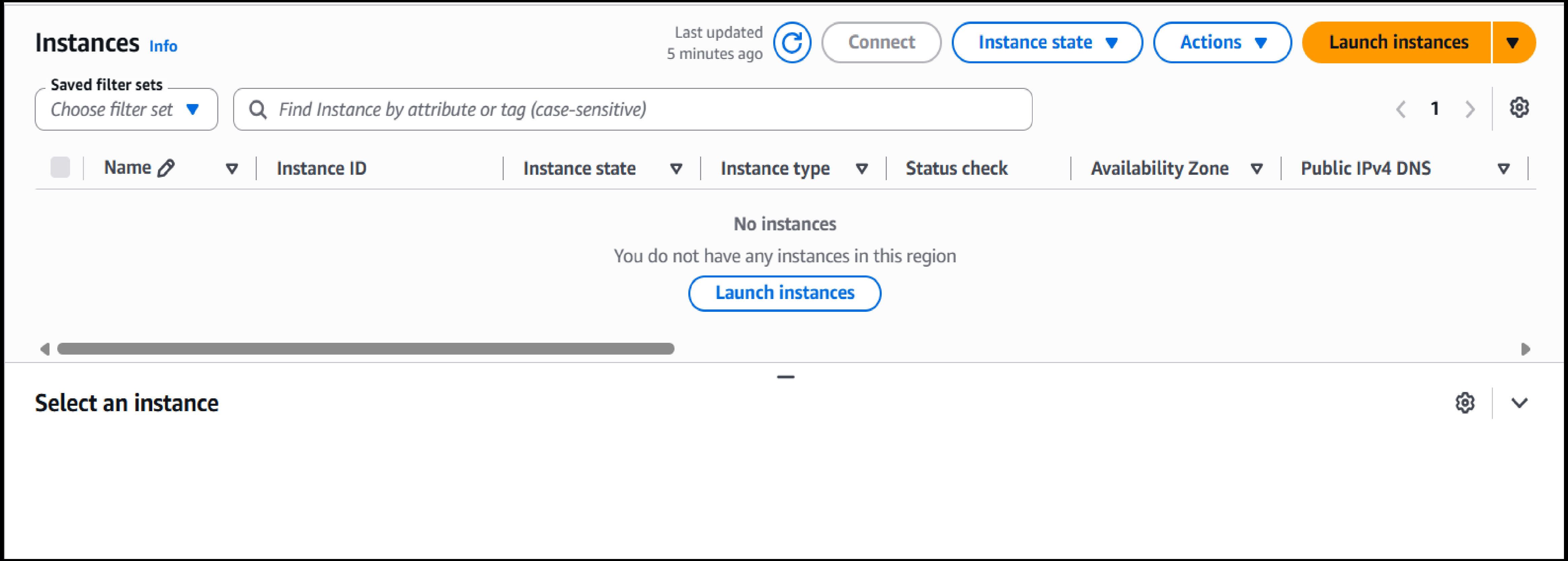


7. Scroll Down and click on Create Security Group

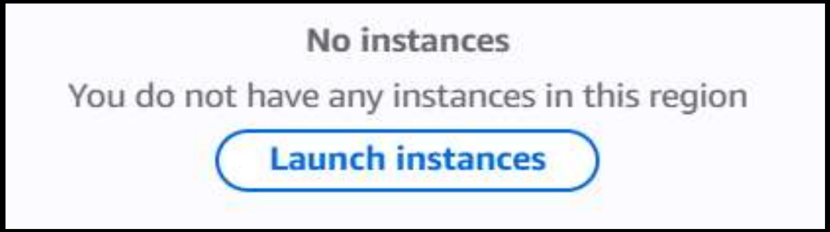


Launch EC2 Instances💻

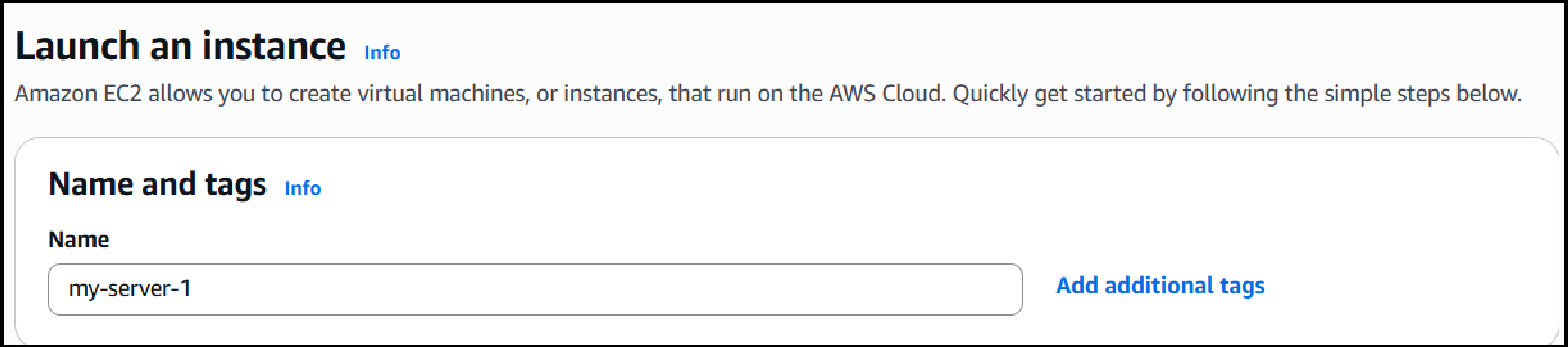
8. Go to the **EC2 Instances Dashboard**

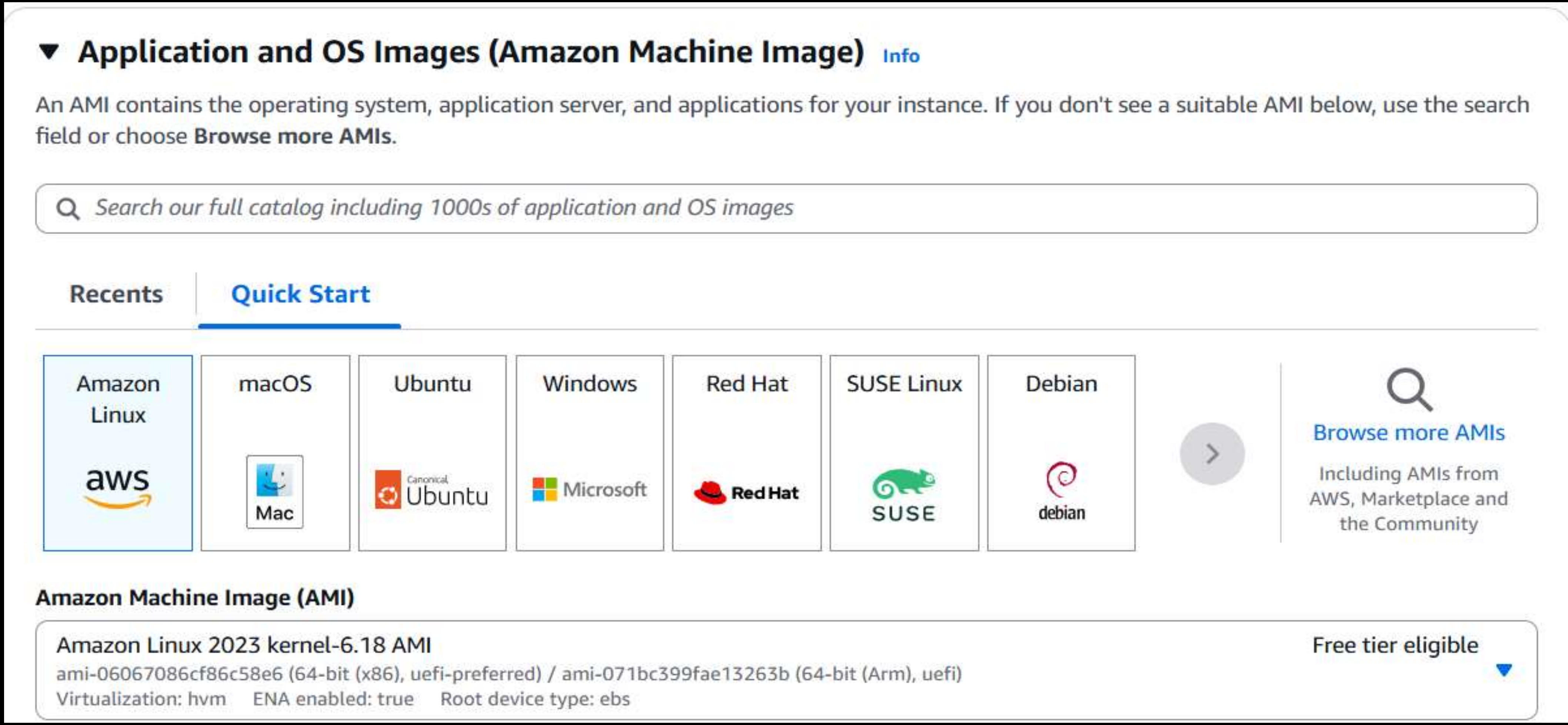


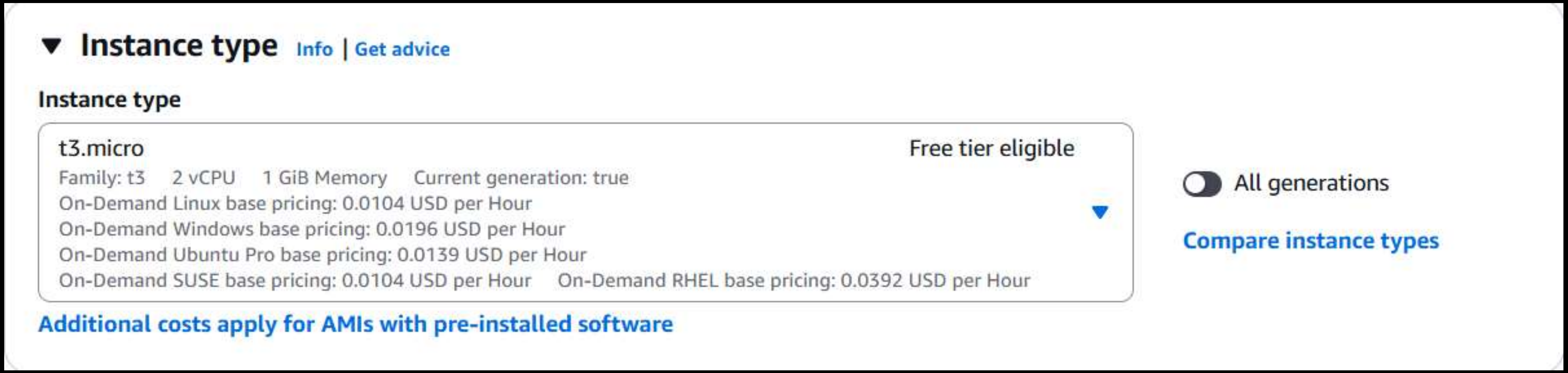
9. Click on Launch Instance



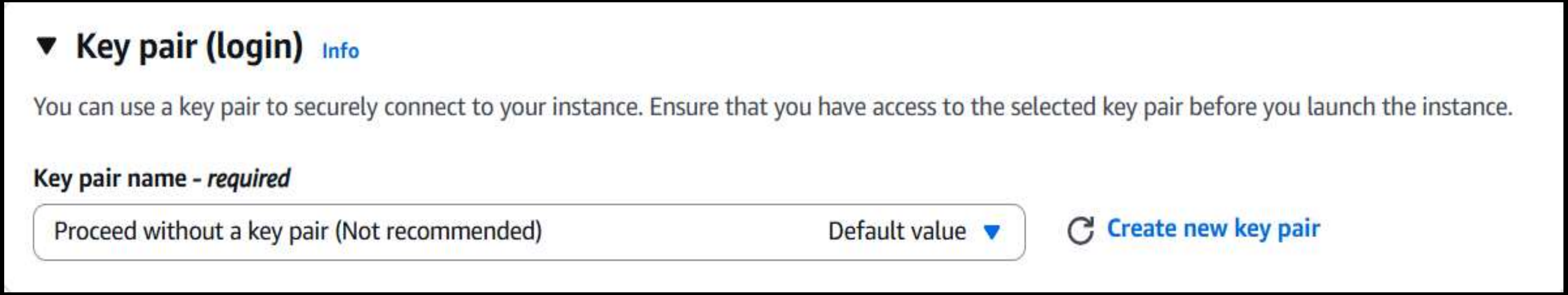
10. Provide a name, Select **AMI** as **Amazon Linux**, and choose **t2.micro** or **t3.micro** (whichever is free-tier eligible).



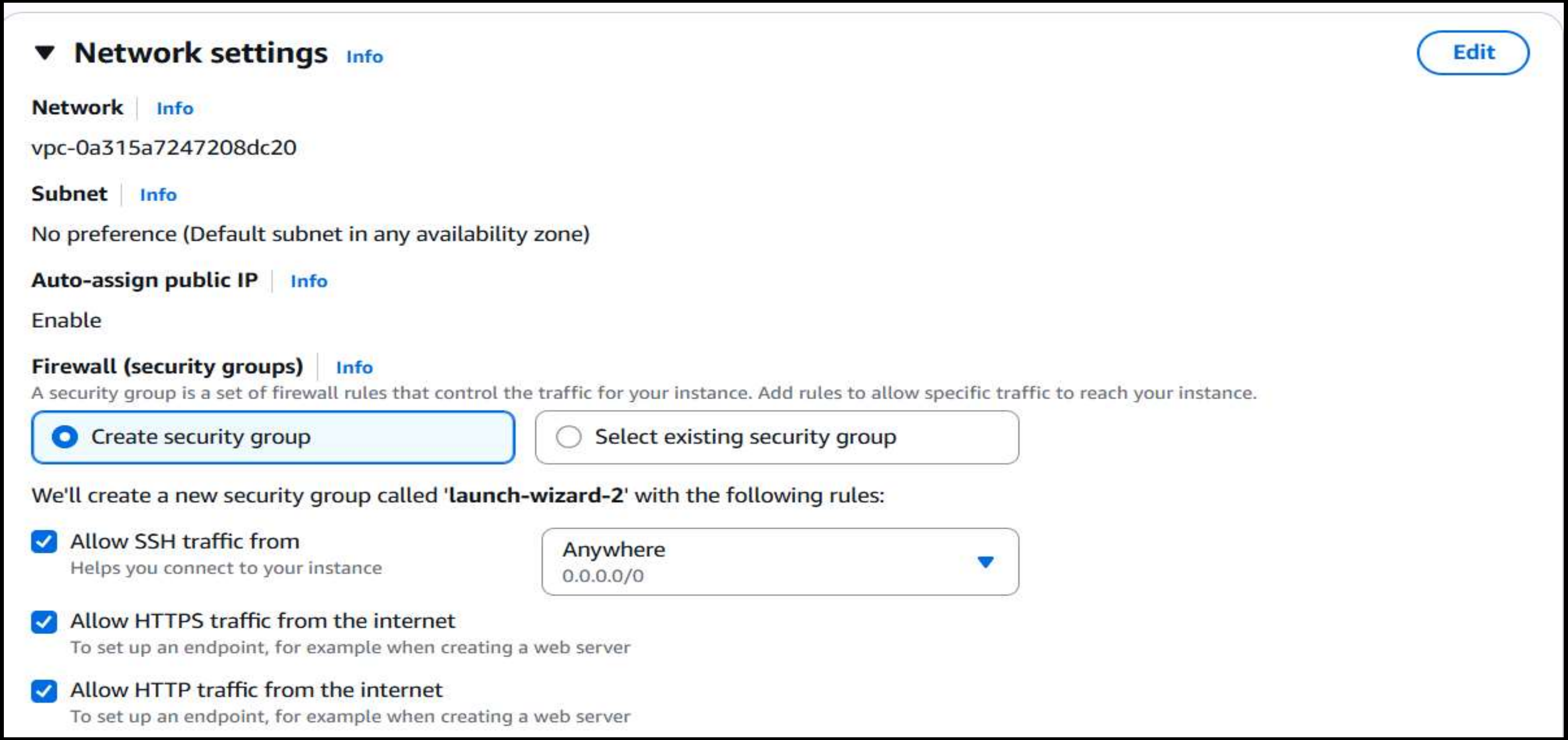

Application and OS Images (Amazon Machine Image)
Info
An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose Browse more AMIs.
Search our full catalog including 1000s of application and OS images
Recents
Quick Start
Amazon Linux
aws
macOS
Mac
Ubuntu
Canonical Ubuntu
Windows
Microsoft
Red Hat
Red Hat
SUSE Linux
SUSE
Debian
debian
Browse more AMIs
Including AMIs from AWS, Marketplace and the Community
Amazon Machine Image (AMI)
Amazon Linux 2023 kernel-6.18 AMI
Free tier eligible
ami-06067086cf86c58e6 (64-bit (x86), uefi-preferred) / ami-071bc399fae13263b (64-bit (Arm), uefi)
Virtualization: hvm
ENA enabled: true
Root device type: ebs



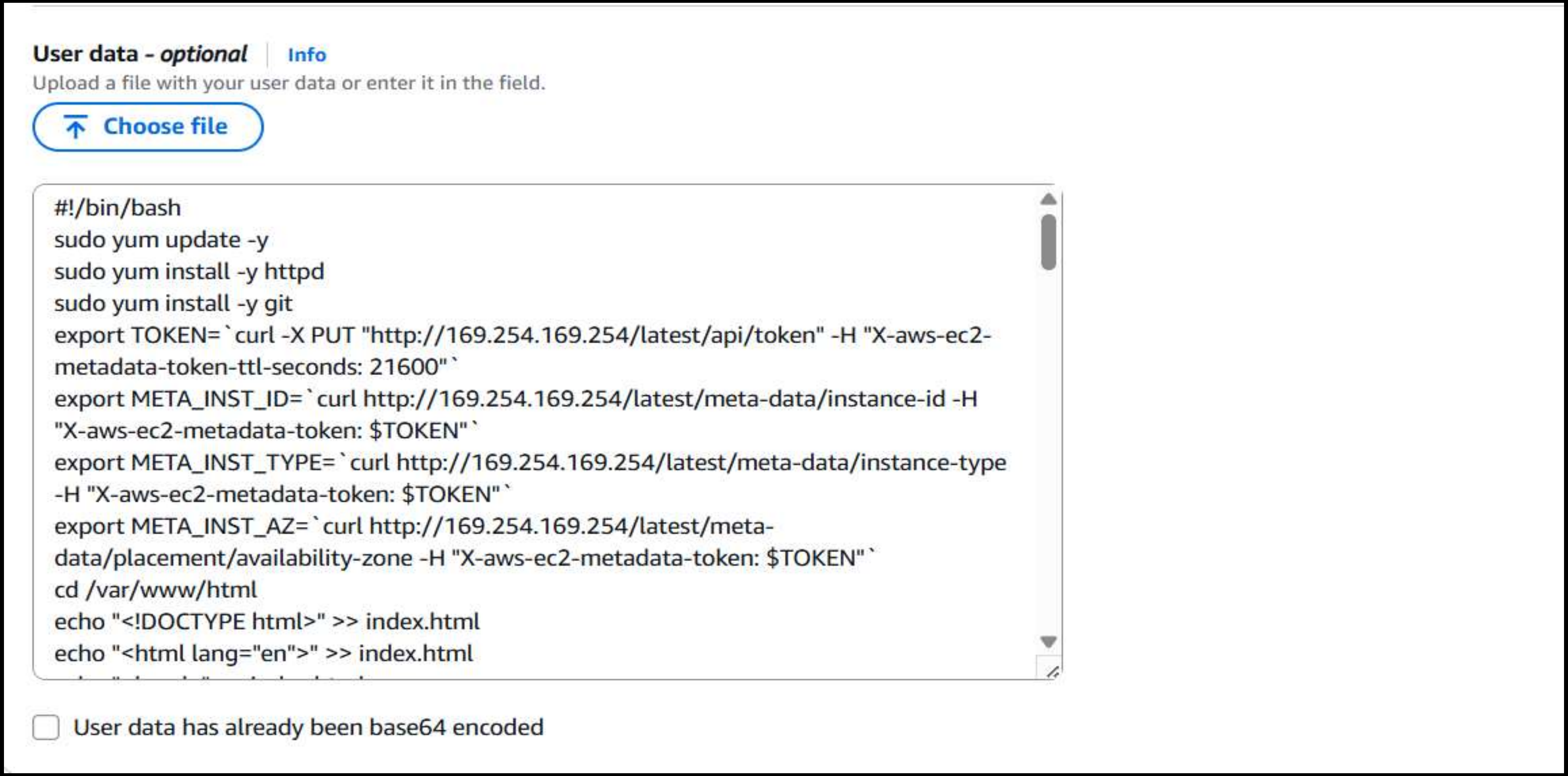
11. Proceed **Without** a key pair:



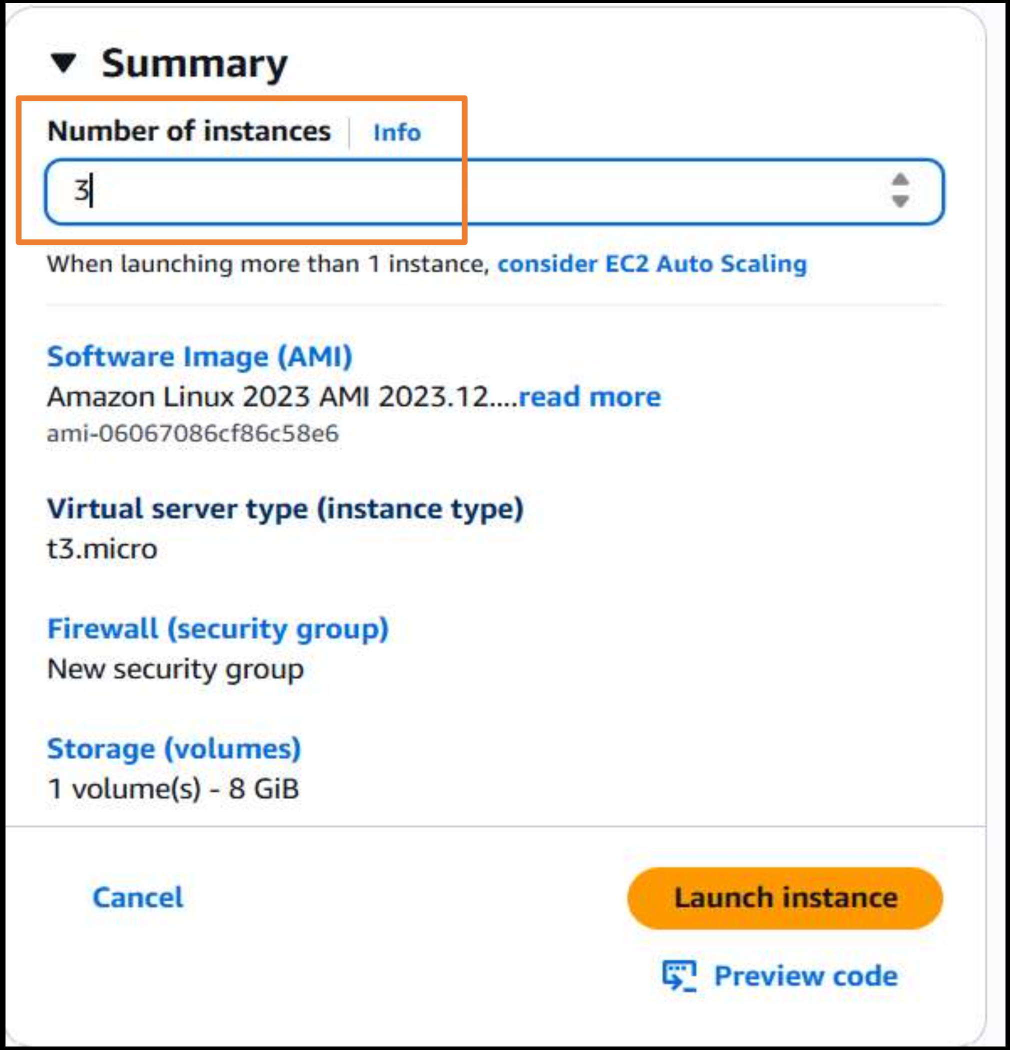
12. In **Networking Setting** , Select the **Create security group** create a earlier.



13. Scroll down to **Advanced Details** and add **user Data.**

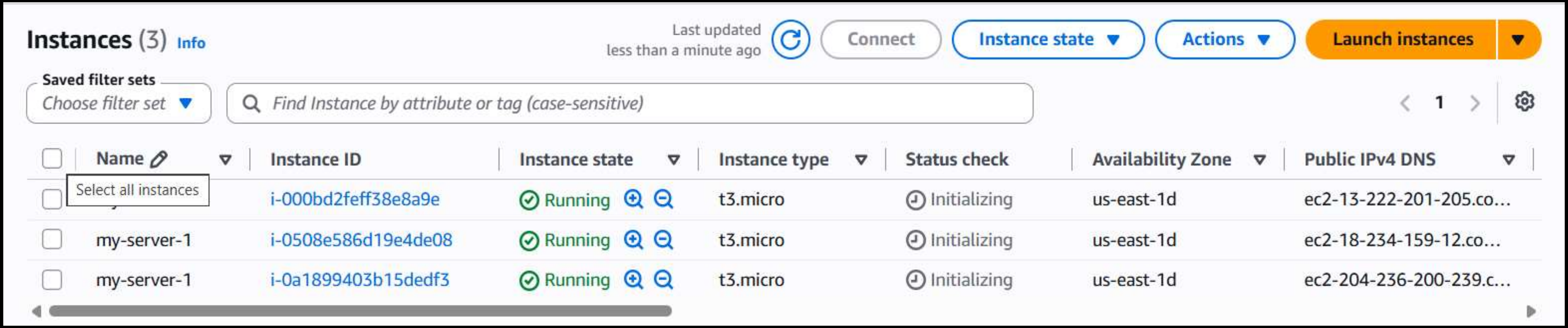


14. Change the **number of instances** from 1 to 3 in the **summary** section.



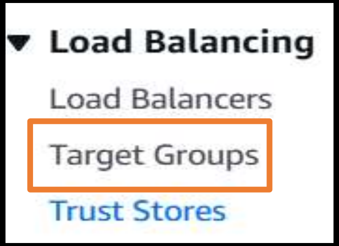
15. Click **Launch Instance**

16. Navigate to the EC2 Instances Dashboard, wait for the 3/3 instance check to pass.

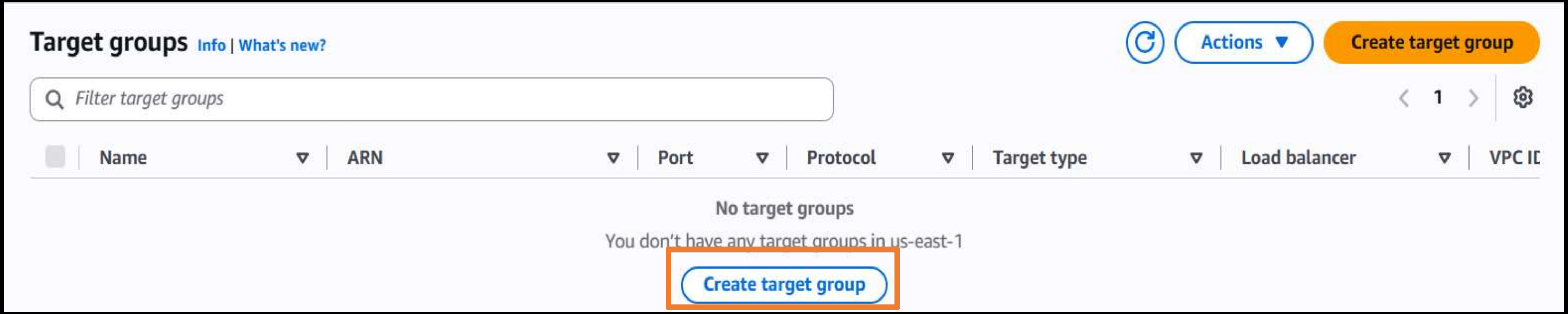


Create the **TARGET GROUPS** 🎯

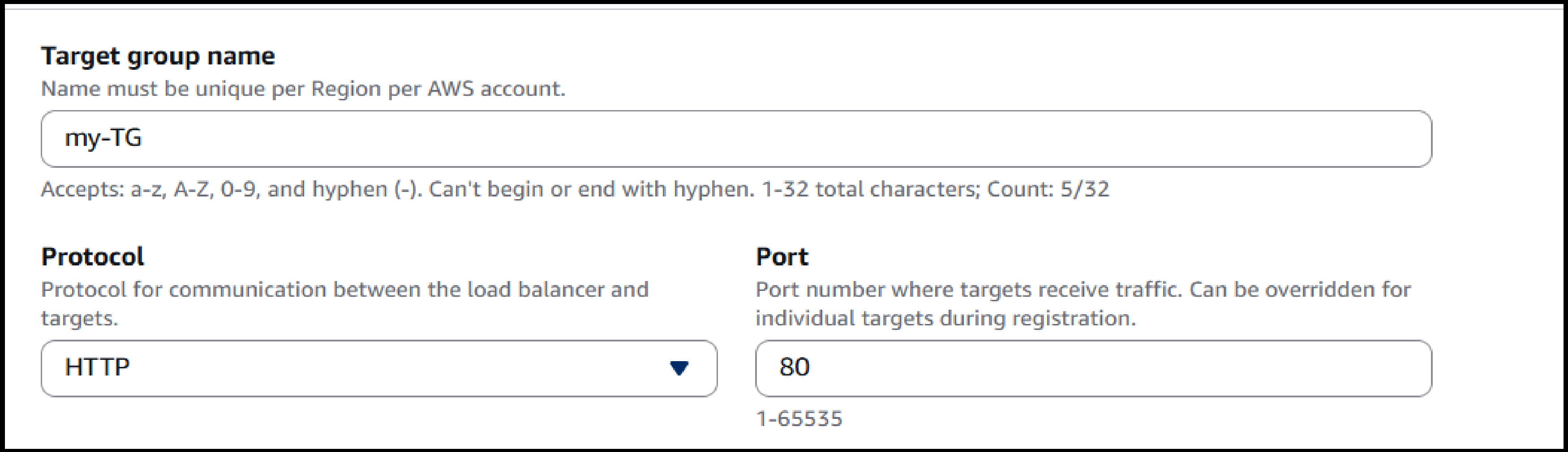
17. In the **Load Balancing** Section, click on **Target Groups.**

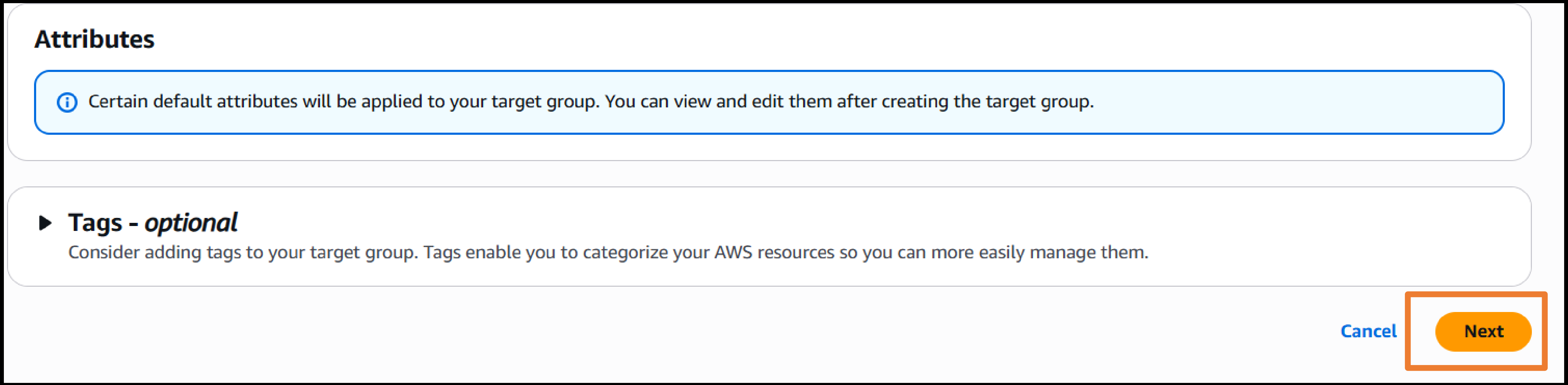


18. Click **Create Target Group**.



19. Enter a **Target Group** name, then click **Next.**





20. Select available instances and Register them.

Register targets - *recommended*

This is an optional step to create a target group. However, to ensure that your load balancer routes traffic to this target group you must register your targets.

Available instances (3/3)

Filter instances

1

	Instance ID	Name	State	Security groups	Zone
☑	i-0beaa14bf23c8cf22	my-server-1	Running	launch-wizard-1	us-east-1d
☑	i-01b5b5f8ad88bd6fb	my-server-1	Running	launch-wizard-1	us-east-1d
☑	i-06e22706e1b1e8771	my-server-1	Running	launch-wizard-1	us-east-1d

3 selected

Ports for the selected instances

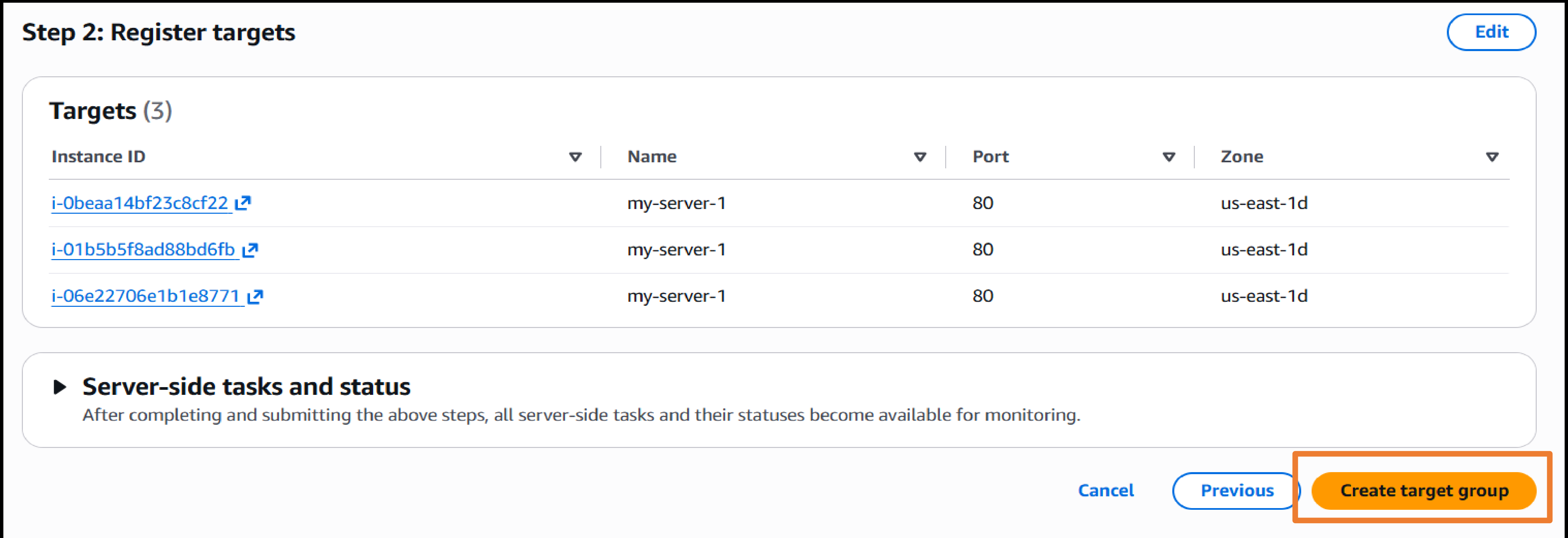
Ports for routing traffic to the selected instances.

80

1-65535 (separate multiple ports with commas)

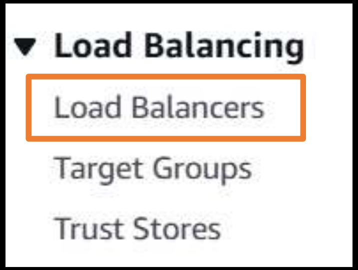
Include as pending below

21. Review Target and click **Create Target Group.**

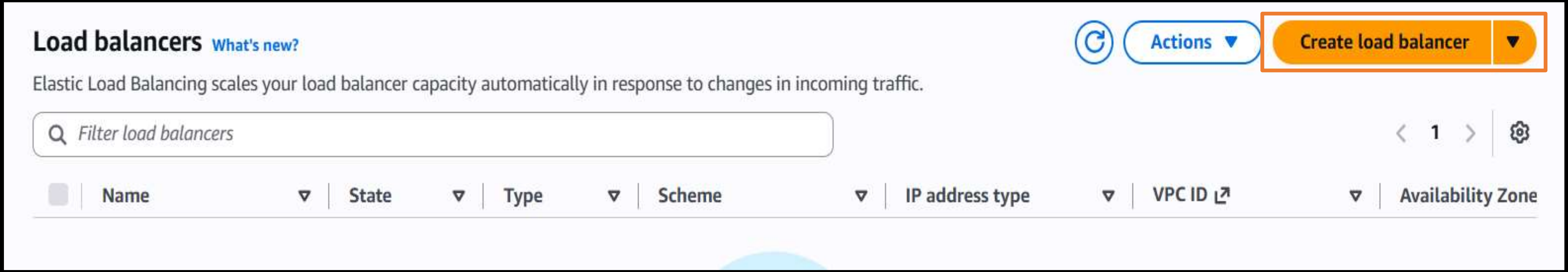


SET UP ELASTIC LOAD BALANCER (ELB)

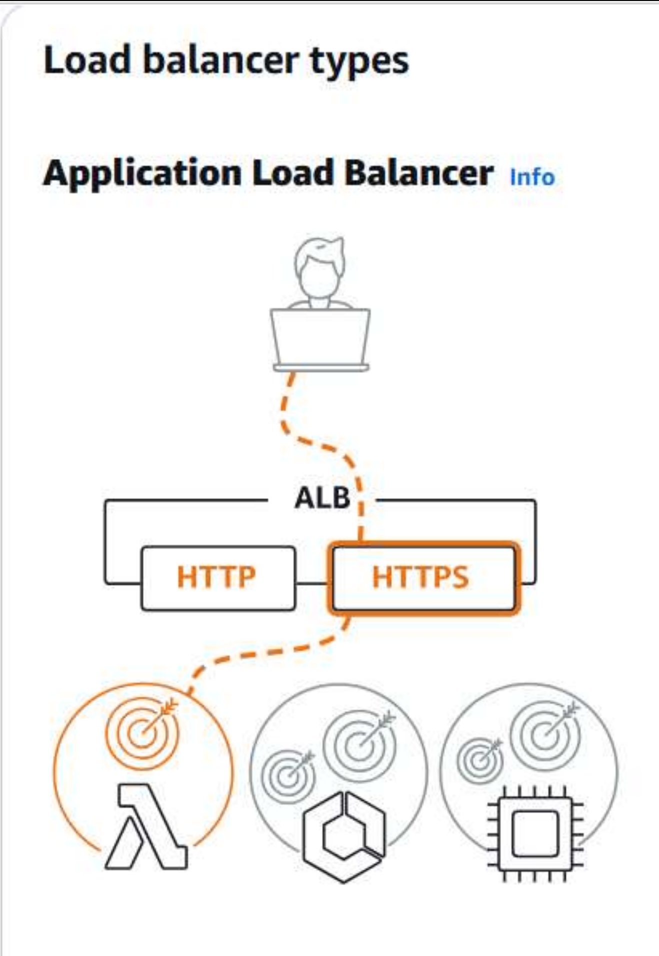
22. Go to **Load Balancers** in the **Load Balancing** sections



23. Click **Create Load Balancer**.



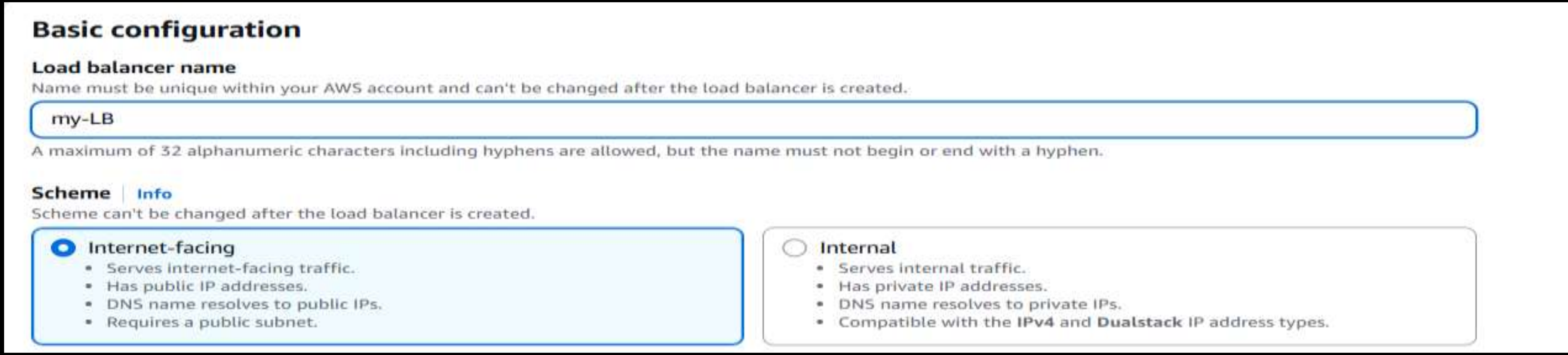
24. Choose the **Application Load Balancer (ALB)** and click **Create**



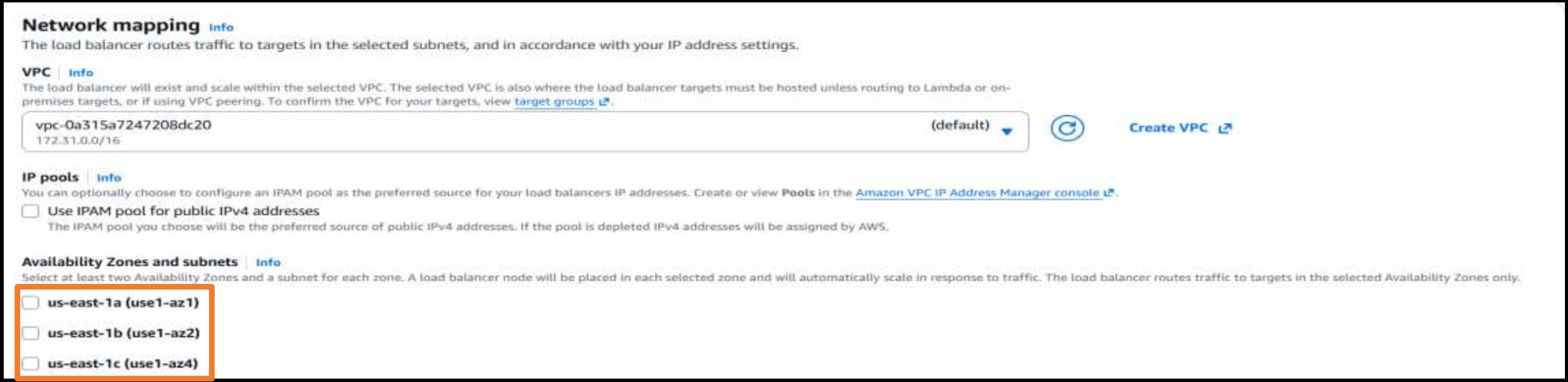
Choose an Application Load Balancer when you need a flexible feature set for your applications with HTTP and HTTPS traffic. Operating at the request level, Application Load Balancers provide advanced routing and visibility features targeted at application architectures, including microservices and containers.

Create

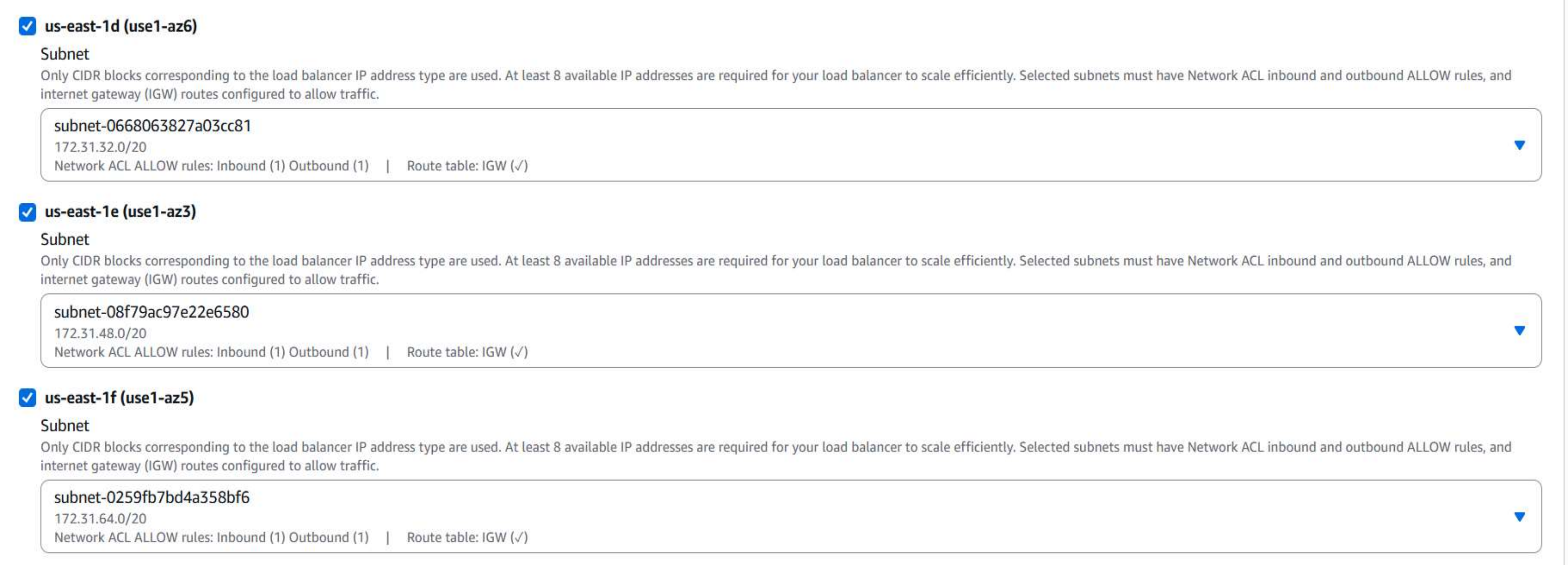
25. Enter the **name** for the Load Balancer.



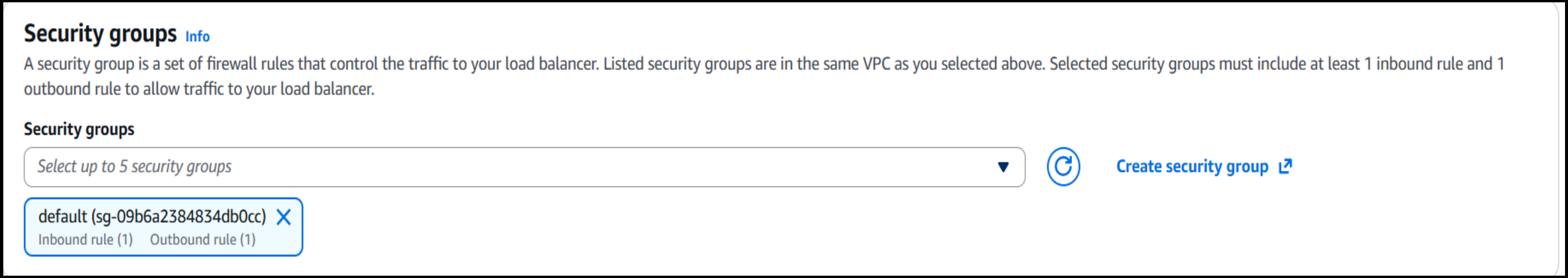
26. In **Network Mapping** choose the **Availability zone.**

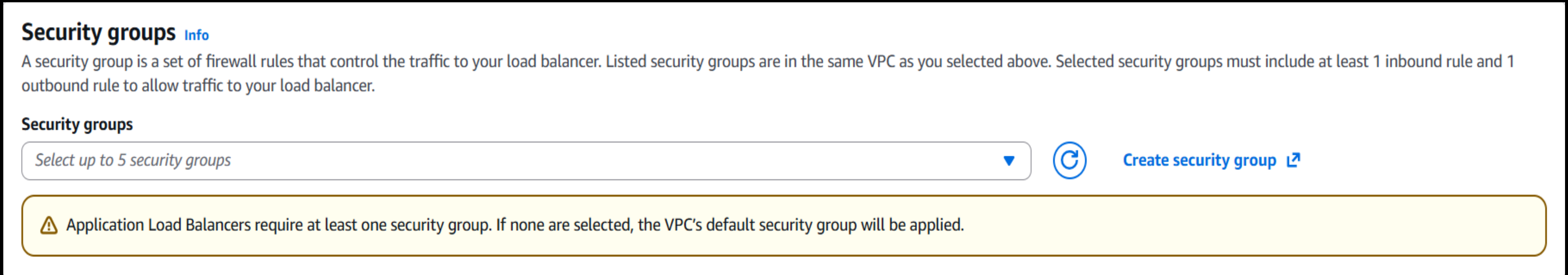


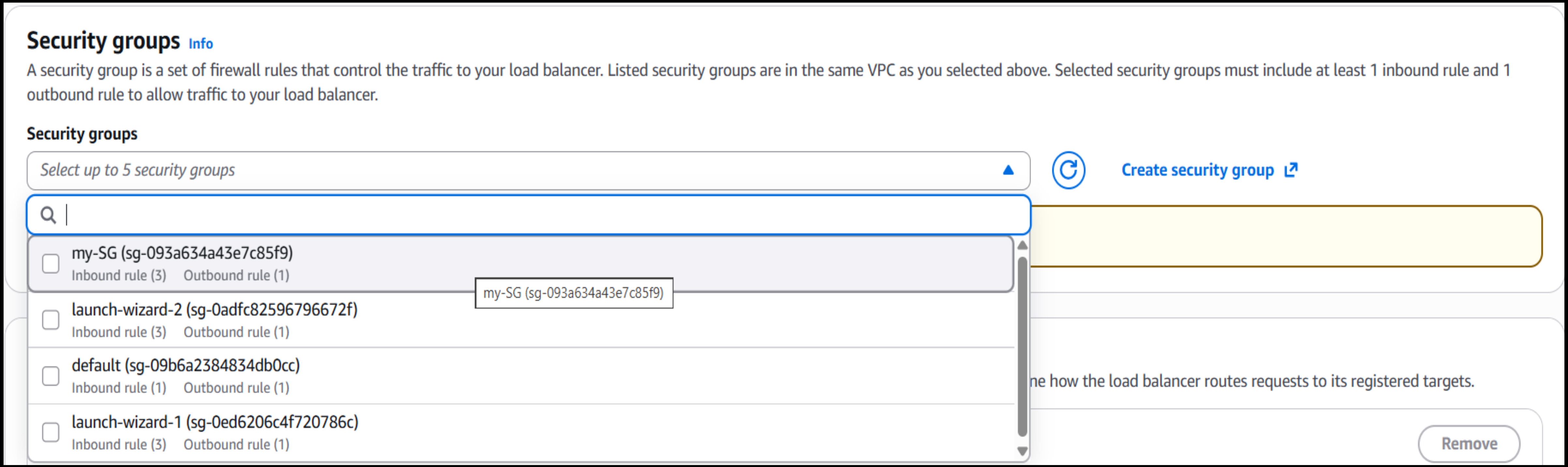
Availability zone

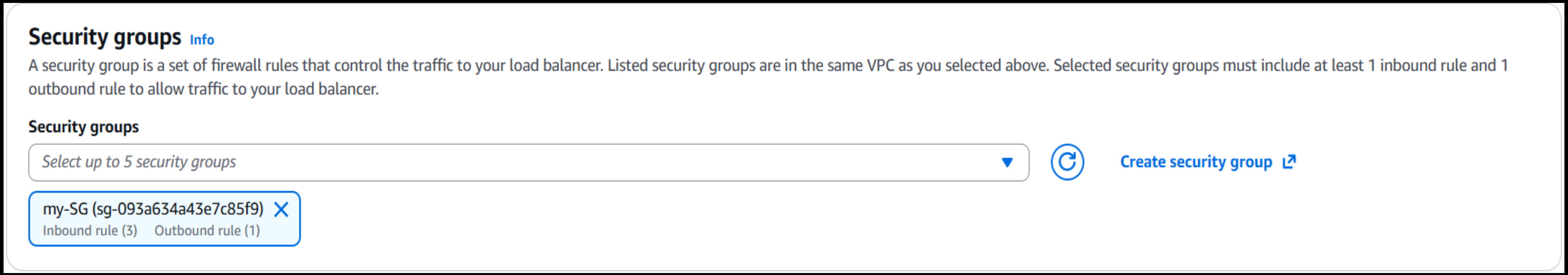


27. In **Security Group** remove the default Security Group and attach the **Security Group created earlier**

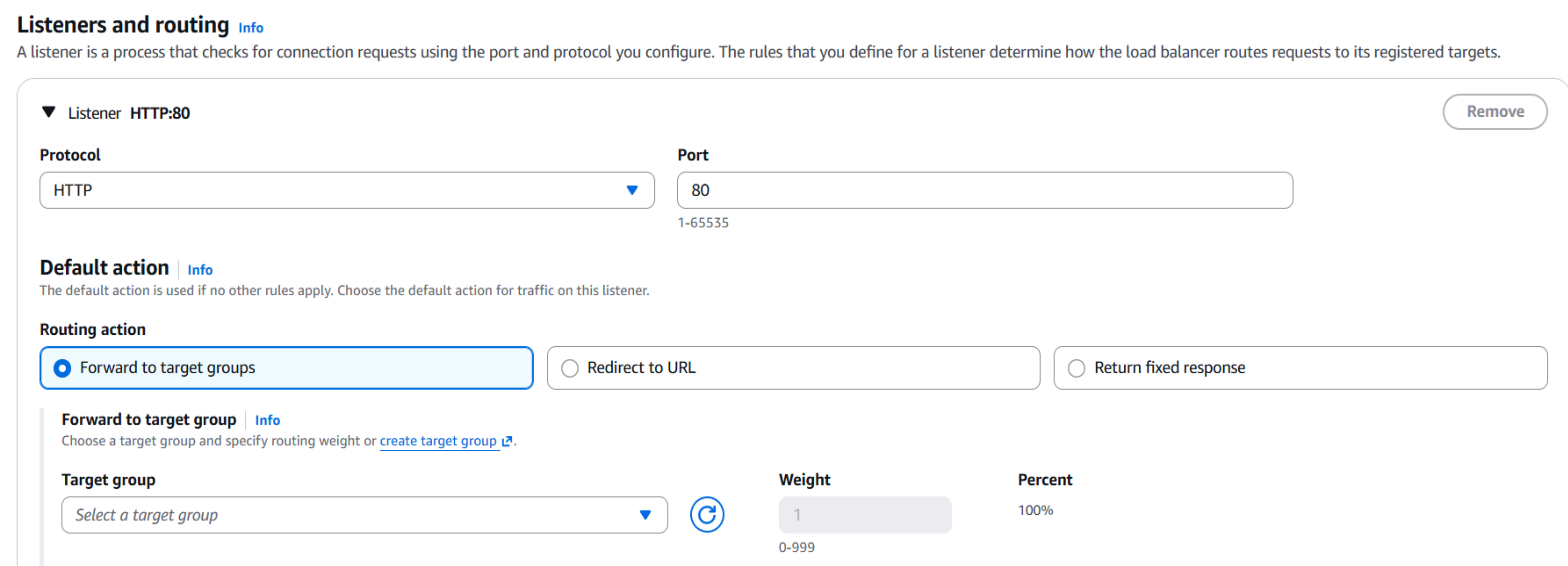


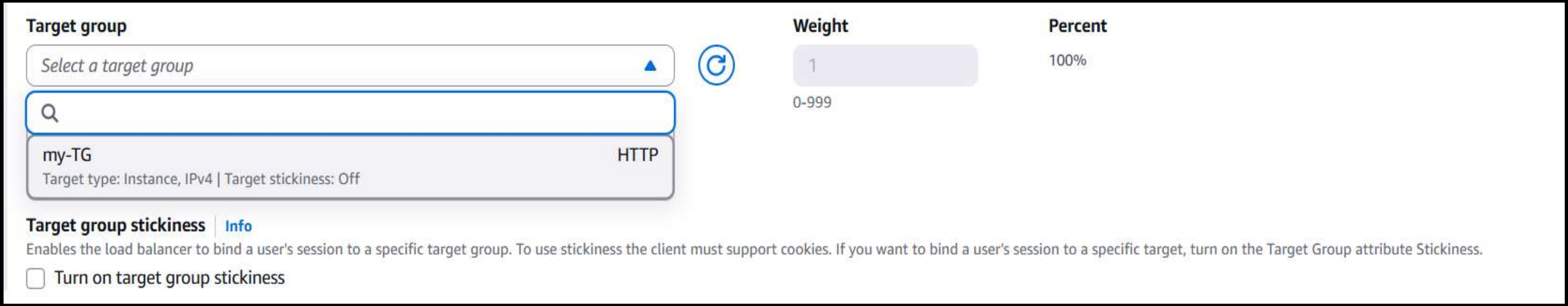



Security groups Info
A security group is a set of firewall rules that control the traffic to your load balancer. Listed security groups are in the same VPC as you selected above. Selected security groups must include at least 1 inbound rule and 1 outbound rule to allow traffic to your load balancer.
Security groups
Select up to 5 security groups
Create security group
my-SG (sg-093a634a43e7c85f9)
Inbound rule (3) Outbound rule (1)
my-SG (sg-093a634a43e7c85f9)
launch-wizard-2 (sg-0adfc82596796672f)
Inbound rule (3) Outbound rule (1)
default (sg-09b6a2384834db0cc)
Inbound rule (1) Outbound rule (1)
launch-wizard-1 (sg-0ed6206c4f720786c)
Inbound rule (3) Outbound rule (1)
ne how the load balancer routes requests to its registered targets.
Remove


Security groups Info
A security group is a set of firewall rules that control the traffic to your load balancer. Listed security groups are in the same VPC as you selected above. Selected security groups must include at least 1 inbound rule and 1 outbound rule to allow traffic to your load balancer.
Security groups
Select up to 5 security groups
Create security group
my-SG (sg-093a634a43e7c85f9)
Inbound rule (3) Outbound rule (1)

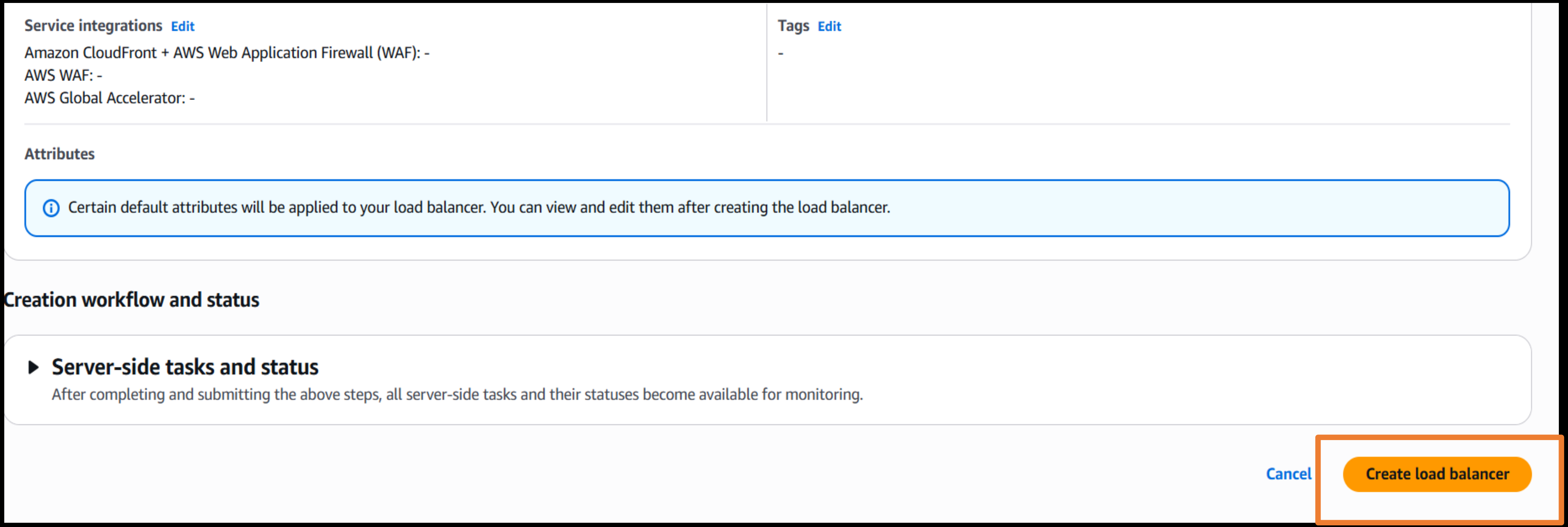
28. In **Listeners and Routing**, attach the **Target Group** created earlier.



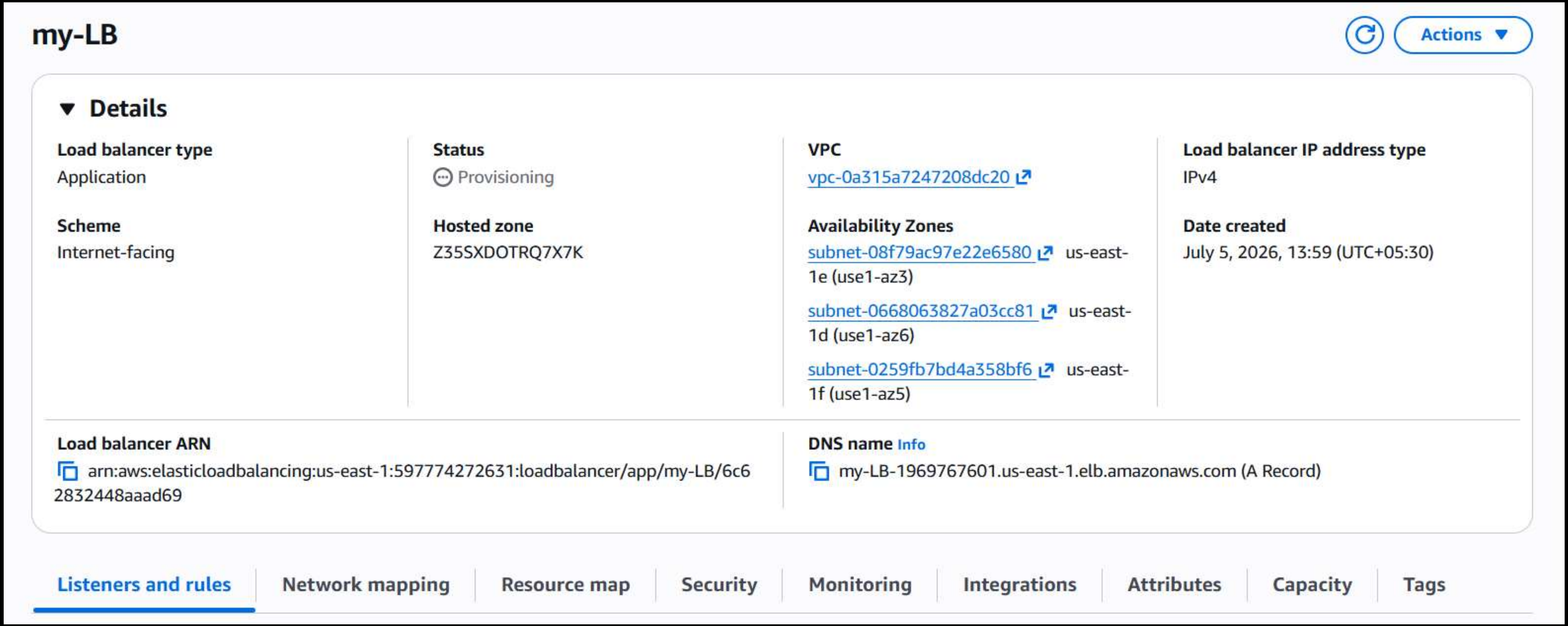

Target group
Select a target group
my-TG
HTTP
Target type: Instance, IPv4 | Target stickiness: Off
Weight
1
0-999
Percent
100%
Target group stickiness
Info
Enables the load balancer to bind a user's session to a specific target group. To use stickiness the client must support cookies. If you want to bind a user's session to a specific target, turn on the Target Group attribute Stickiness.
Turn on target group stickiness

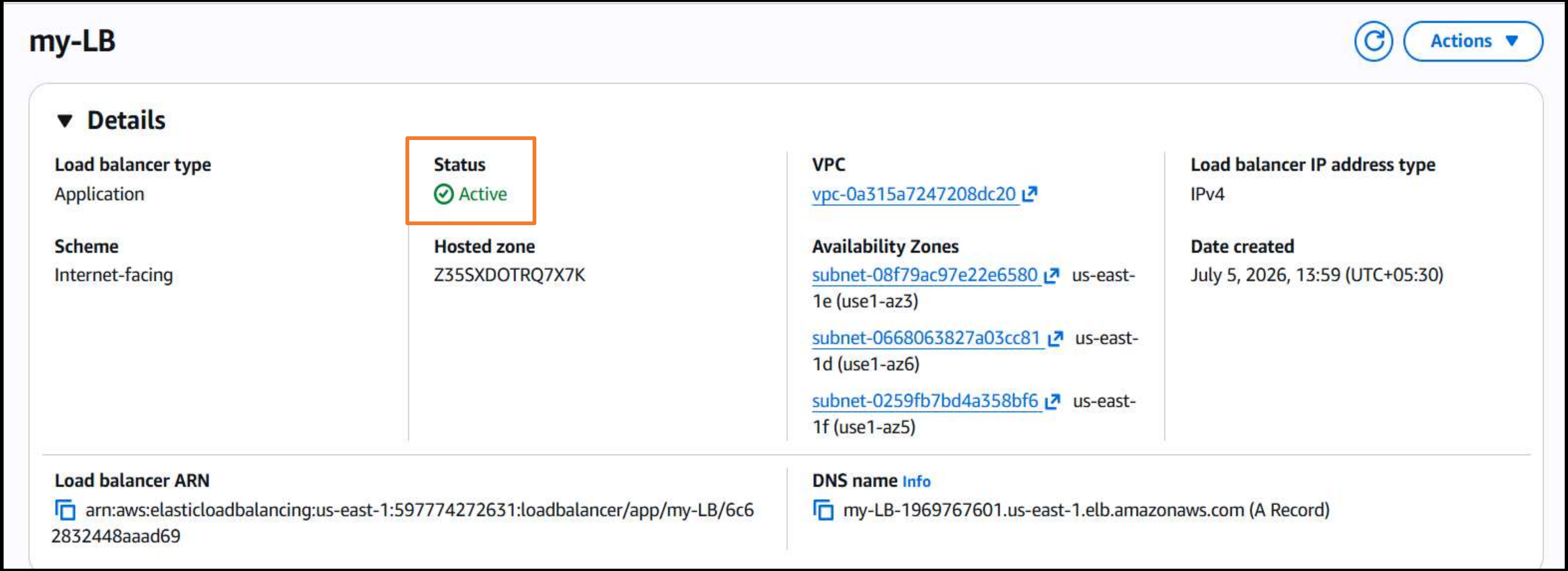

Target group
my-TG
HTTP
Target type: Instance, IPv4 | Target stickiness: Off
Weight
1
0-999
Percent
100%

29. Scroll down and click **Create Load Balancer.**

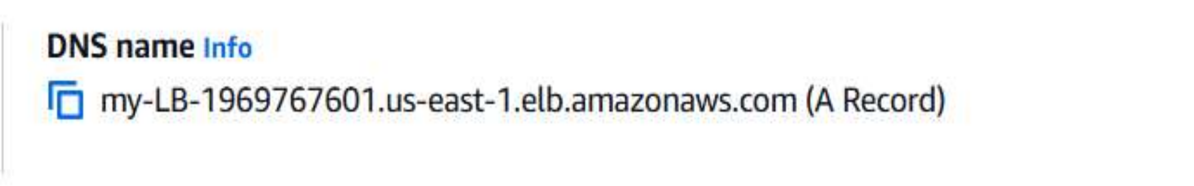


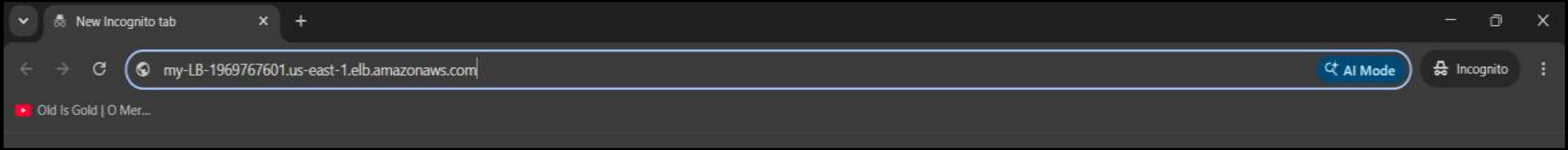
30. Wait for the **ELB status** to become **Active**

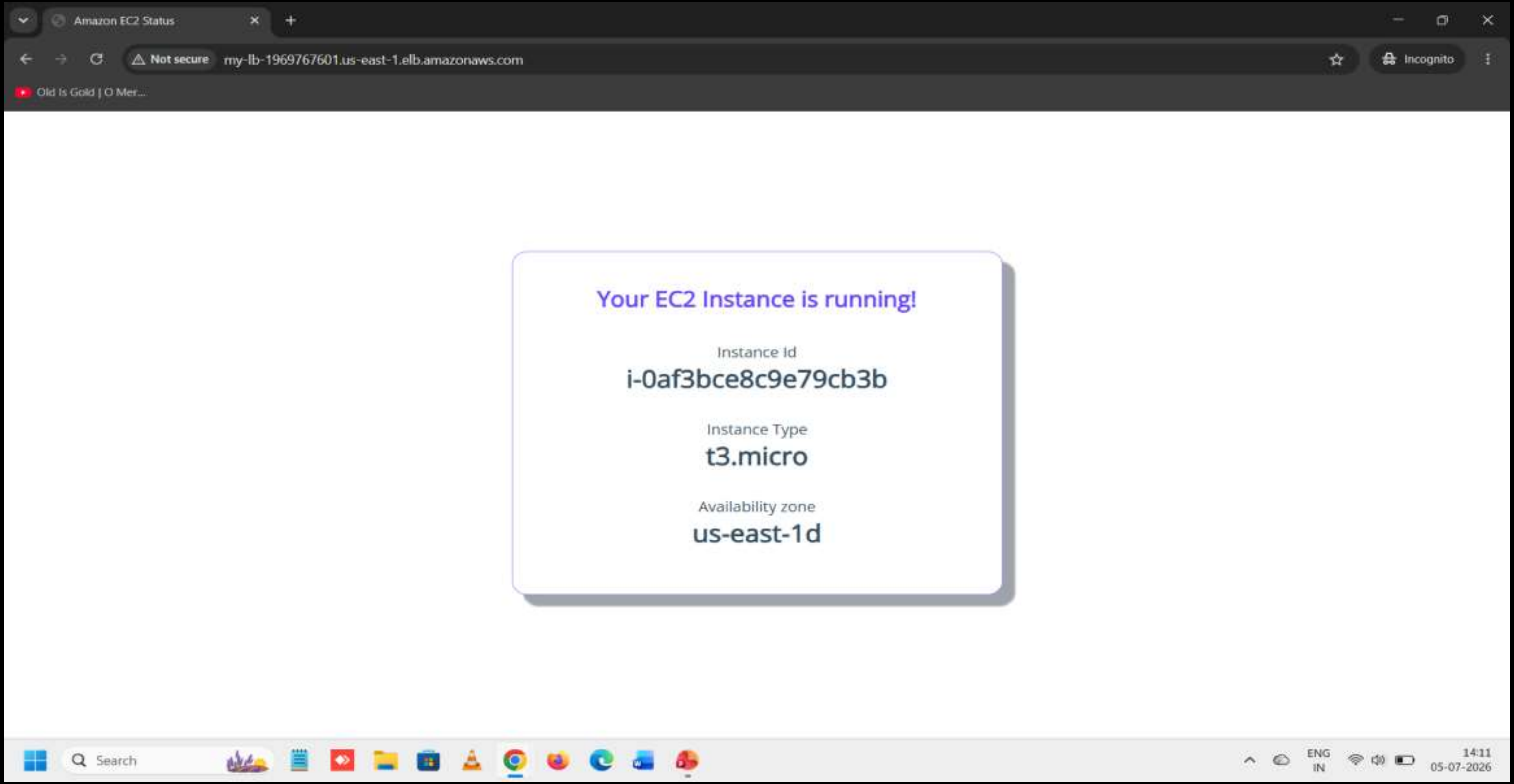



my-LB
Actions
Details
Load balancer type
Application
Status
Active
VPC
vpc-0a315a7247208dc20
Load balancer IP address type
IPv4
Scheme
Internet-facing
Hosted zone
Z35SXDOTRQ7X7K
Availability Zones
subnet-08f79ac97e22e6580 us-east-1e (use1-az3)
subnet-0668063827a03cc81 us-east-1d (use1-az6)
subnet-0259fb7bd4a358bf6 us-east-1f (use1-az5)
Date created
July 5, 2026, 13:59 (UTC+05:30)
Load balancer ARN
arn:aws:elasticloadbalancing:us-east-1:597774272631:loadbalancer/app/my-LB/6c62832448aaad69
DNS name Info
my-LB-1969767601.us-east-1.elb.amazonaws.com (A Record)

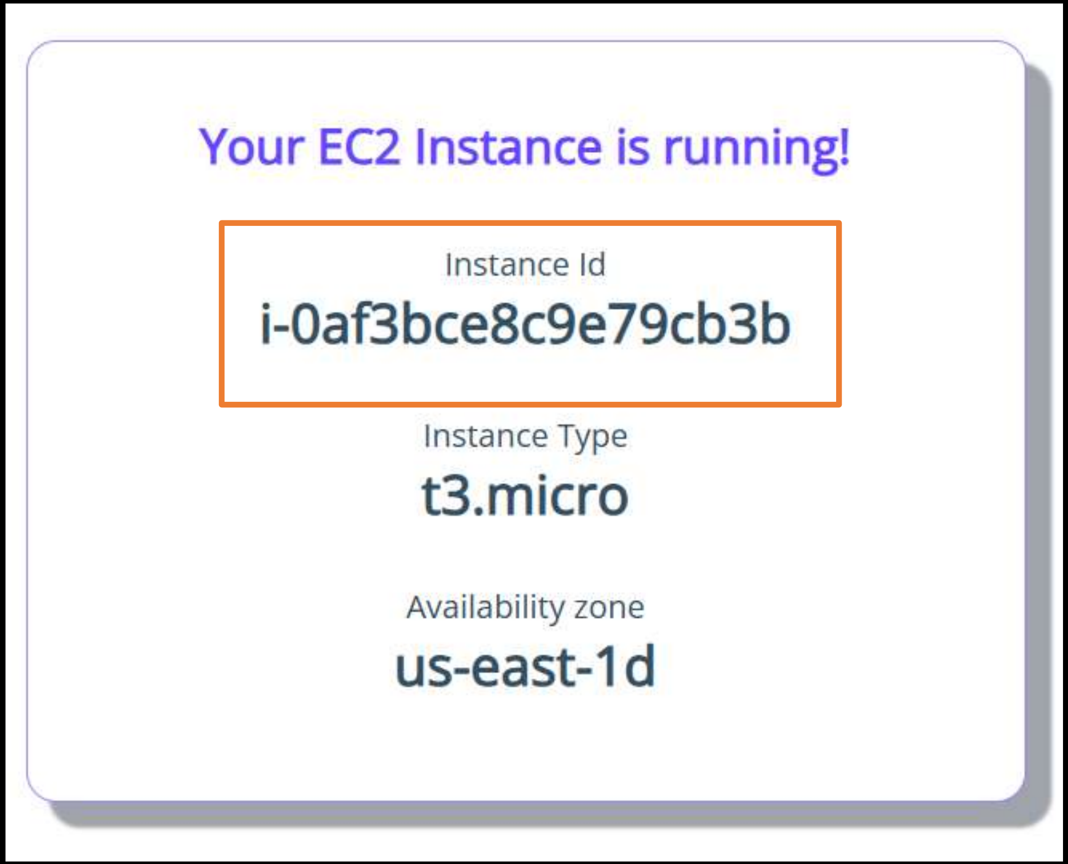
31. Copy the **DNS mode** And Go to the Incognito window and paste **DNS mode.**

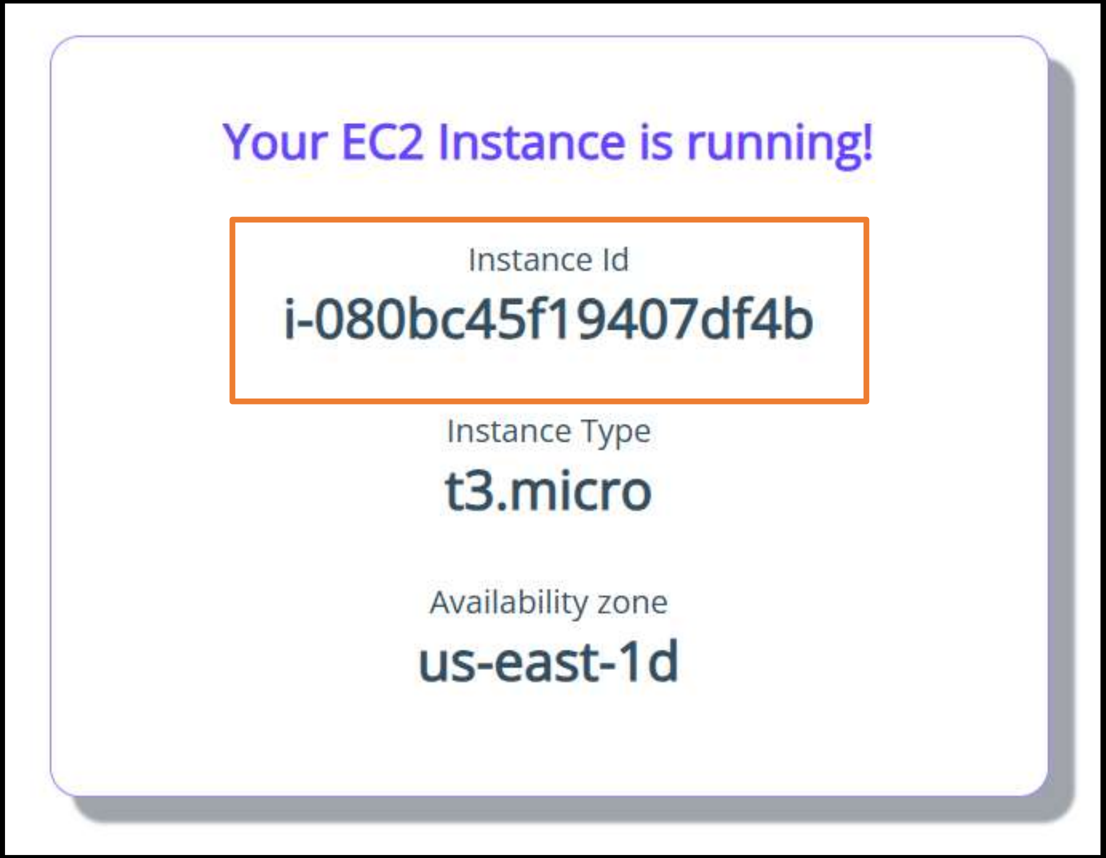





Amazon EC2 Status
Not secure
my-lb-1969767601.us-east-1.elb.amazonaws.com
Incognito
Old Is Gold | O Mer...
Your EC2 Instance is running!
Instance Id
i-0af3bce8c9e79cb3b
Instance Type
t3.micro
Availability zone
us-east-1d
Search
ENG
IN
14:11
05-07-2026

32. Click **Refresh Button.** And see the **Instance rotate.**





Architecture

Users
|
▼
Load Balancer
|
▼
Target Group
┌──────┼──────┐
▼ ▼ ▼
EC2-1 EC2-2 EC2-3
(Healthy)(Healthy)(Unhealthy)
▲ ▲
| |
Traffic sent only to healthy instances

Security Group protects:
✓ Load Balancer
✓ EC2 Instances

Thank You!

YOUR SUPPORT MEANS A LOT